

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE ALL SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

1 of 6

ASSEMBLY BY		All dimensions in millimetres
		All torques in Kilogram metres

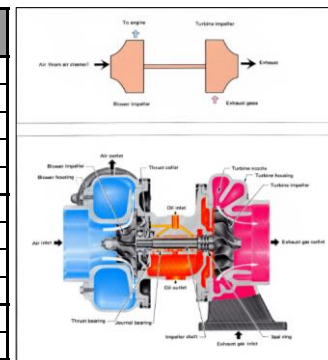
Notes :

- Before assembly parts, wash and clean the parts, allow them to dry by air blow, and apply engine oil to the sliding parts.
- Remove burrs completely from the parts (Pay attention to seal ring grooves, in particular).
- When reassembling the rotor assembly after the dynamic balance check has been made, be sure to align all the match marks stamped on the component parts (Turbine Shaft ; Nut ; Blower Impeller ; Flinger and Collar).

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	SPV	
1	Turbine Rotor	Seal ring Groove	Fit seal ring into Turbine rotor. Apply a coat of engine oil (SAE #30) to the side of seal ring and install the seal ring in such a way that the gap of each ring is positioned at 90° apart from that of the neighboring ring or 45° apart from oil filler port on the center housing.					

Thickness of Seal Ring

	Standard size		Tolerance Width	Repair Limit Width	Actual
KTR 100	Turbine Side	2,2	-0,08	1,85	
			-0,1		
	Blower Side	1,95	-0,08	1,85	
			-0,1		
		4	-	-	
KTR 110	Turbine Side	2,2	-0,08	1,85	
			-0,1		
	Blower Side	1,95	-0,08	1,45	
			-0,1		
KTR 130		3	-0,08	2,85	
			-0,1		
KTR 150		3	-0,08	2,85	
			-0,1		



No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
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2	Shield		Install shield into the Turbine Rotor					
3	Center Housing	Seal ring seat	Install center housing on the Turbine When installing the center housing, take care to prevent the seal ring from slipping to installation					
4	Journal Bearing	Journal bearing surface	Install journal bearing Apply sufficient amount of engine oil to the inside, outside and both sides of the journal Bearing prior to installation					

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	MACHINE MODEL		FINISH DATE / TIME		PAGE	

Measurement of Journal Bearing, Center Housing & Wheel Shaft

2 of 6

	Standard Size	Tolerance		Repair limit		Actual	
		Shaft	Hole	Shaft	Hole	Shaft	Hole
KTR 100							
Outside diameter of journal bearing	25	-0,04	0,02	24,91	25,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	17	-0,03	0,02	16,94	17,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 110							
Outside diameter of journal bearing	25	-0,04	0,02	24,91	25,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	17	-0,03	0,02	16,94	17,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 130							
Outside diameter of journal bearing	30	-0,04	0,02	29,91	30,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	20	-0,03	0,02	19,94	20,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 150							
Outside diameter of journal bearing	30	-0.04	0.02	29,91	30,04		
Inner diameter of center housing		-0.06	0				
Inner diameter of Journal bearing	20	-0.03	0.02	19,94	20,05		
Outside diameter of wheel shaft		-0.05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

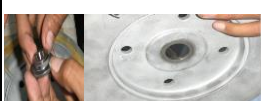
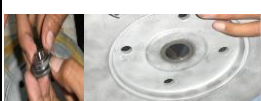
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5	Thrust		- Install thrust collar - Make sure that the thrust collar is free from foreign material on both sides, apply engine oil (SAE #30) both sides of the collar, and install that collar with the match mark aligning with marking mark stamp on the turbine shaft. - Drive a dowel pin in the center housing.					
6	Thrust Bearing		- Install thrust bearing with the dowel pin fitted in position - Before installing the trust Bearing, apply engine oil (SAE #30) to both sides of the oil hole					

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	MACHINE MODEL		FINISH DATE / TIME		PAGE	

Thickness of thrust bearing

3 of 6

KTR 100	Standard size	Tolerance Width	Repair limit Width	Actual Width
	5	-0,08 -0,11	4,86	
KTR 110	Standard size	Tolerance Width	Repair limit Width	Actual Width
	5	-0,08 -0,11	4,86	
KTR 130	Standard size	Tolerance Width	Repair limit Width	Actual Width
	6	-0,08 -0,11	5,86	
KTR 150	Standard size	Tolerance Width	Repair limit Width	Actual Width
	6	-0,08 -0,11	5,86	

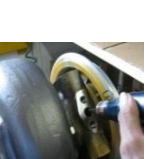
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7	Flinger	One seal ring	Fit seal ring into flinger					  
			Apply engine oil to the sides of the seal ring					
			Install flinger into insert (back plate)					
			Let the gap of the seal ring on the flinger					
			Direct to the oil filler port on the center housing					
8	Insert (Back Plate)	Two seal ring	Let the gap of the seal ring in such away that the gap of each ring is positioned at 90° apart from that of the Neighboring ring or at 45° apart from oil filler port on the center housing.					
		For KTR 100 , KTR 130	Fit an O-ring and install insert					
			Tighten the insert mounting bolts and then bend the lock washer					
			Act. KTR 100 = kgm	KTR 100	1.25 ± 0.25			
			Act. KTR 130 = kgm	KTR 130	3.15 ± 0.35			
		For KTR 110, KTR 150	Fit an O-ring on housing center					 
			Install back plate onto Turbine Rotor until contact the center housing and make sure that flinger with the match mark aligning with the marking mark stamp on the turbine shaft.					
			Tighten the back plate mounting bolts and bend the lock washer.					
			Act. KTR 110 = kgm	KTR 110	3.15 ± 0.35			
			Act. KTR 150 = kgm	KTR 150	6.75 ± 0.75			

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9	Blower Impellers		Install blowers impellers onto the turbine rotor in the following manner immerse blower impeller in the oil (SAE #30) that is heated at 140° to 160° C for 5 to 20 minutes or immerse Blower					       
			Impellers with hot air gun for 5-10 minutes.					
			Install the blower impellers into the turbine rotor, aligning the match mark with the match mark stamp on the turbine shaft.					
			Apply engine oil to the inner face of the blower impeller and install the impeller without using excessive force					
			When tightening the nut, take care not allow any match marks to be out of alignment. Make sure that nut with the match mark aligning with the marking mark stamp on the turbine shaft and blower impeller.					
			Tightening torque :					
			Act. KTR 100 = kgm	KTR 100	4.25 ± 0.25			
			Act. KTR 110 = kgm	KTR 110	4.5 ± 0.25			
			Act. KTR 130 = kgm	KTR 130	8.5 ± 0.5			
			Act. KTR 150 = kgm	KTR 150	9.5 ± 0.5			
			Check End Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.06 to 0.11 0,18		
			Actual KTR 110 : = mm	KTR 110	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 130 : = mm	KTR 130	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 150 : = mm	KTR 150	Standard Repair limit	0.08 to 0.13 0,18		
			Check Radial Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.20 to 0.44 0,6		
			Actual KTR 110 : = mm	KTR 110	Standard Repair limit	0.25 to 0.44 0,6		
			Actual KTR 130 : = mm	KTR 130	Standard Repair limit	0.25 to 0.43 0,6		
			Actual KTR 150 : = mm	KTR 150	Standard Repair limit	0.25 to 0.43 0,6		

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	MACHINE MODEL		FINISH DATE / TIME		PAGE	

5 of 6

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10	Housing	Turbine Housing	Take care not to damage the blower impeller and turbine impeller when installing the blower housing and turbine housing					   	
			For KTR 110 , KTR 150 Install the Turbine housing into the center housing and the rotor assembly. Apply a thin coat of anti seiz compound to the flange section of turbine housing. Install lock plate, tighten mounting bolt. Apply a thin coat of anti seiz compound to the thread of bolt.						
			Tighten mounting bolt :						
			Act. KTR 110 = kgm	KTR 110	5.0 ± 0.5				
			Act. KTR 150 = kgm	KTR150	5.75 ± 0.25				
		Blower Impeller housing	Install blower housing, aligning it with the center housing and the rotor assembly. Apply a thin coat of Grease to the flange section.					 	
			For KTR 100 , KTR 130 Install the blower housing and turbine housing into the center housing and the rotor assembly.						
			Install V-Band clamp and tighten then torque of the V-Band locknut.						
			Act. KTR 100 = kgm	KTR 100	0.9 ± 0.1				
			Act. KTR 130 = kgm	KTR 130	0.9 ± 0.1				
			Install V-Band clamp and tighten then torque of the V-Band locknut.					 	
			Act. KTR 110 = kgm	KTR 110	0.9 ± 0.1				
			Act. KTR 150 = kgm	KTR150	0.9 ± 0.1				
			Measure the clearance in the radial direction in the following manner.					 	
			Depress the end of the rotor in the radial direction with a finger. Measure the clearance between the blower impeller and the housing by using filler gauge. Tolerance (Min.) :						
			Act. KTR 100 = mm	KTR 100	0,15				
			Act. KTR 110 = mm	KTR 110	0,2				
			Act. KTR 130 = mm	KTR 130	0,2				
			Act. KTR 150 = mm	KTR 150	0,2				
AFTER COMPLETING THE ASSEMBLY,ROTATE THE ROTOR ASSEMBLY WHILE DEPRESSING IT LIGTHLY AND MAKE SURE THAT THE ROTOR CAN BE ROTATED SMOOTHLY WITHOUT BINDING.									

AFTER COMPLETING THE ASSEMBLY, ROTATE THE ROTOR ASSEMBLY WHILE DEPRESSING IT LIGHTLY AND MAKE SURE THAT THE ROTOR CAN BE ROTATED SMOOTHLY WITHOUT BINDING.

